Avinash Amanchi

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Education

Emerald High School — Dublin, CA
High School Diploma — Expected June 2027

GPA: 3.87 weighted cumulative; 4.13 weighted (grades 10–12). SAT: 1500 (750 Math, 750 Reading & Writing)
Honors: National Spanish Exam Gold Medal (top 5% nationally, Level 4); Presidential Athlete Award (2x); Scholar Athlete Award (2x)
Relevant Coursework: AP Computer Science A, AP Computer Science Principles, AP Chemistry, AP English Language & Composition, Calculus, Honors Spanish 4

Las Positas College — Livermore, CA
Dual Enrollment — Jun 2023 – Present

Computing Fundamentals I (CS 1) and Introduction to Statistics (Fall 2026); Geometry (Summer 2023)

University of Notre Dame Pre-College (Online) — Notre Dame, IN
Business: From Startup to Success — Mar 2024 – Apr 2024

Compared strategy at Lululemon, Nike, and AT&T; built a startup plan whose pricing work carried into EAA Detailing

Research & Technical Projects

Sepsis Vitals — Sepsis Early-Warning Model — GitHub
Founder & Lead Developer — 2026 – Present

- Built a sepsis early-warning system with FHIR data ingestion, Sepsis-3 labeling, a calibrated risk model, drift and fairness monitoring, and a 546-test suite
- Ran a vitals-only ablation on a 3,000-patient held-out synthetic cohort: AUROC fell only from 0.916 to 0.858, supporting use where lab results are unavailable
- Completed CITI human-subjects training and earned PhysioNet credentialed access to move the model from synthetic data onto MIMIC, a de-identified ICU database

Wordless — Accessibility Tool for People with Aphasia — Dublin, CA
Co-Creator & Developer — 2025 – Present

- Built the pipeline that reads a user's account history and matches a few fragmented words to the likely support issue, returning a one-tap confirmation instead of requiring full sentences
- Placed 3rd of 60+ teams at the OpenAI x Start2 x Zendesk Hackathon after building and demoing live in 4 hours; won $10K in OpenAI credits
- Continued development after the hackathon and earned local press coverage; integrating Wordless into BeeLine Reader's accessibility platform with founder Nick Lum

FieldCraft — Voice-First Invoicing for Tradespeople — GitHub | Live Site
Founder & Lead Developer — 2025 – 2026

- Built a voice-first tool that lets tradespeople speak a job description and receive a finished invoice
- Recruited 12 detailers, landscapers, and cleaners for a 6-week pilot and shipped fixes from their feedback, cutting median invoice time from 8 minutes to 2
- Retained 7 companies as active users after the pilot

AI-Trader — Algorithmic Trading Assistant — GitHub
Lead Developer — 2025 – 2026

- Designed an AI trading assistant around safety controls: a human approval gate on every order, a deterministic risk engine, position limits, audit logs, and a kill switch
- Integrated the Anthropic API via MCP with Alpaca and read-only Robinhood access; tested on paper accounts
- Put the codebase through a security audit (276 passing tests) and fixed the CSRF and XSS vulnerabilities it identified

S.A.G.E. — Accessibility-First App for Older Adults — GitHub | Live Site
Co-Creator & UX Research Lead — 2023 – 2024

- Co-designed an app for older adults with large touch targets, high contrast, and simplified navigation, built with Python, Java, and MIT App Inventor
- Ran usability testing with 50 seniors at community centers and rebuilt the interface around observed behavior, raising task completion 35%, cutting onboarding time 40%, and reaching 90% positive feedback

Additional Projects

- **ConvoAutopsy** — AI app that maps patterns and breakdowns in conversations (GitHub | Live Site)
- **AI Resume Analyzer** — Python app that parses resumes and simulates ATS scoring (GitHub | Live Site)
- **Personal Finance Tracker** — JavaScript budgeting app with interactive dashboards (GitHub | Live Site)

Work Experience

EAA Detailing | Website — Dublin, CA
Co-Founder & Operations Manager — 2024 – Present

- Co-founded a student-run auto detailing business at 16 with two partners, serving Dublin, Pleasanton, Livermore, and San Ramon
- Grew bookings from 5 to 20+ per month through self-run ads and social media; served 100+ clients (60%+ repeat) for five-figure revenue
- Set pricing, built the booking website, and manage supplies, scheduling, and customer communication; detail vehicles most weekends

City of Dublin — The Wave Aquatic Center — Dublin, CA
Aquatics Safety Attendant — Summers 2024 – 2025

- Ran the slide tower solo on weekends, dispatching 150+ riders per hour while enforcing height and safety rules
- Radioed lifeguards on hazards, resolved guest complaints, and trained new attendants after returning for a second season

Lawrence Berkeley National Laboratory — Berkeley, CA
Research Intern — May 2023 – Aug 2023

- Wrote Python scripts to process root and leaf measurement data for a plant-growth study
- Structured raw observations for the analysis pipeline, documented methodology, and presented findings to researchers

Leadership & Activities

Neorons — India (Remote)
Co-Founder & Program Operations Lead — 2024 – Present

- Co-founded a youth mental-health organization in India, where psychiatric care is scarce; run operations from California
- Coordinated 8 events across 6 states, including suicide-prevention hackathons, AI training for 140 teachers, and a championship for 84 blind girl athletes

Dublin Unified School District Financial Literacy Curriculum Council — Dublin, CA
Student Representative — Mar 2023 – 2026

- Surveyed 180+ students to demonstrate demand for a personal finance course in a district that had none
- Evaluated 4 curriculum platforms against national standards and co-wrote the semester course framework
- Advocated to administrators as the council's only student member until the elective was adopted; it now runs at both district high schools

Words for the World — Bay Area Chapter — Dublin, CA
Co-Founder, Secretary & Marketing Outreach Director — Feb 2023 – Present

- Grew the chapter to 40 members and ran 6 letter-writing events drawing 50+ students
- Wrote and edited 200+ letters for cancer wards and shelters
- Recruited and launched a New York chapter; training incoming officers for both chapters

Computer Science Honor Society, Emerald High School — Dublin, CA
Co-President — 2025 – 2026

- Grew the chapter from 12 to 25 members in 4 months by pitching CS at club fairs to students new to the field
- Ran weekly meetings, organized 3 hackathons, and set up group projects so members built real products

Varsity Soccer, Emerald High School — Dublin, CA
Team Captain & Goalkeeper — 2023 – Present

Skills & Interests

- **Technical**: Python, Java, JavaScript, HTML/CSS, React, Vite, Git/GitHub, Bash, FHIR, Anthropic API/MCP, Excel
- **Certifications**: CITI Human Subjects Research; American Red Cross CPR/AED for Professional Rescuers, First Aid, Emergency Oxygen
- **Language**: English (fluent), Spanish (intermediate), Telugu (conversational)
- **Interests**: Long bike rides and hikes through the Danville and Mt. Diablo hills